

The slide features a dark background with several decorative circles. In the top-left corner, there is a thin purple circle. In the top-right corner, there are two overlapping circles with a yellow-to-blue gradient. On the right side, there is a large circle with a blue-to-purple gradient. In the bottom-left corner, there is a thin purple circle and a larger circle with a yellow-to-blue gradient. In the bottom-right corner, there is a large black circle with a thin purple outline.

Apresentação Trabalho GA - Redes 2

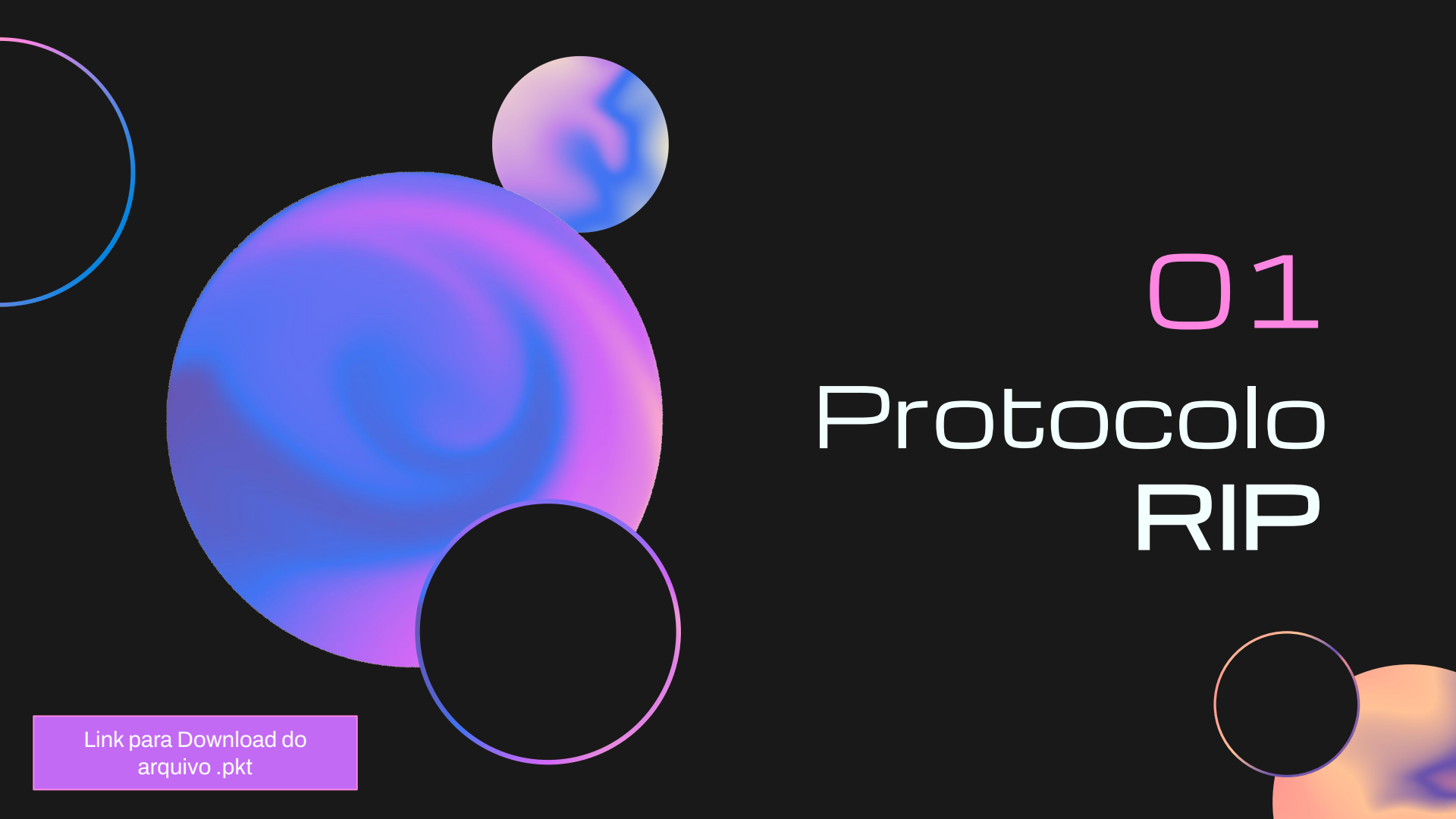
Protocolo de Roteamento

Estudante: Vítor Soares Souza

SOFTWARE UTILIZADO

Simulação do roteamento feito no
Cisco Packet Tracer



The background is dark with several abstract elements: a large blue-to-purple gradient sphere in the center-left, a smaller blue-to-purple gradient sphere above it, a thin blue circle on the top left, a thin purple circle at the bottom center, and a thin orange circle on the bottom right. The text '01' is in a light blue color.

01

Protocolo RIP

Link para Download do
arquivo .pkt

Configurações

Topologia

<i>3 roteadores</i>	<i>3 Computadores</i>	<i>1 Switch</i>
---------------------	-----------------------	-----------------

Foram criadas 5 redes

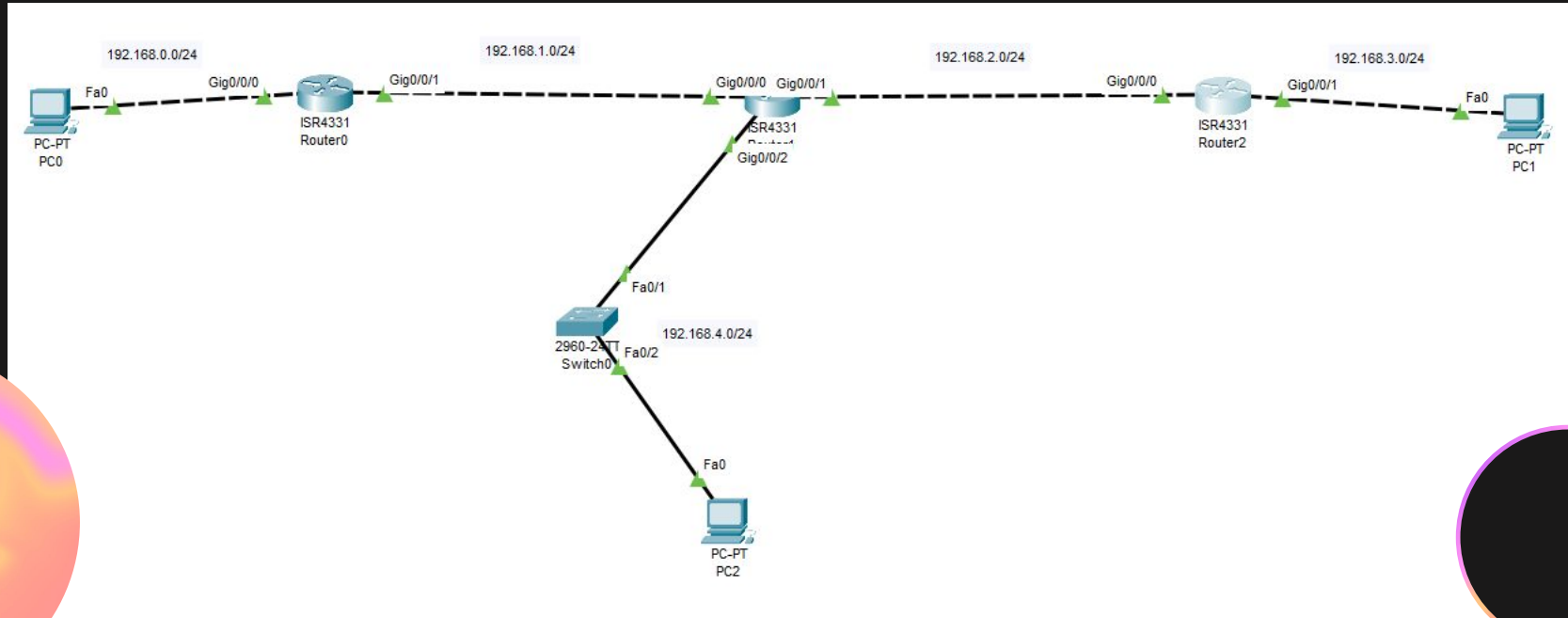
<i>192.168.0.0/24</i>	<i>192.168.1.0/24</i>	<i>192.168.2.0/24</i>	<i>192.168.3.0/24</i>	<i>192.168.4.0/24</i>
-----------------------	-----------------------	-----------------------	-----------------------	-----------------------

```

Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 8 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface        Send Recv Triggered RIP Key-chain
  GigabitEthernet0/0/0  12 1
  GigabitEthernet0/0/1  12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.0.0
  192.168.1.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.1.2      120          00:00:21
Distance: (default is 120)
    
```

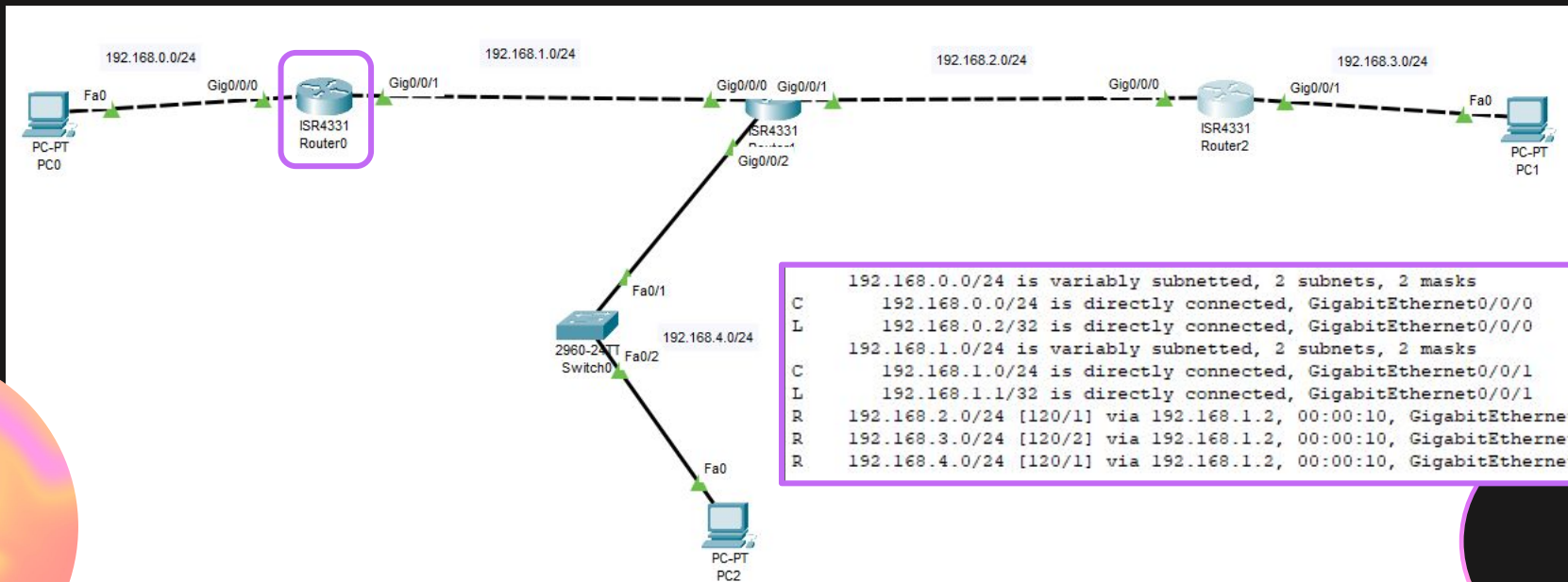
- Distance default é 120
- << "Show IP protocols" no roteador 0
- Se baseia em contagens de saltos para determinar a melhor rota.

Topologia

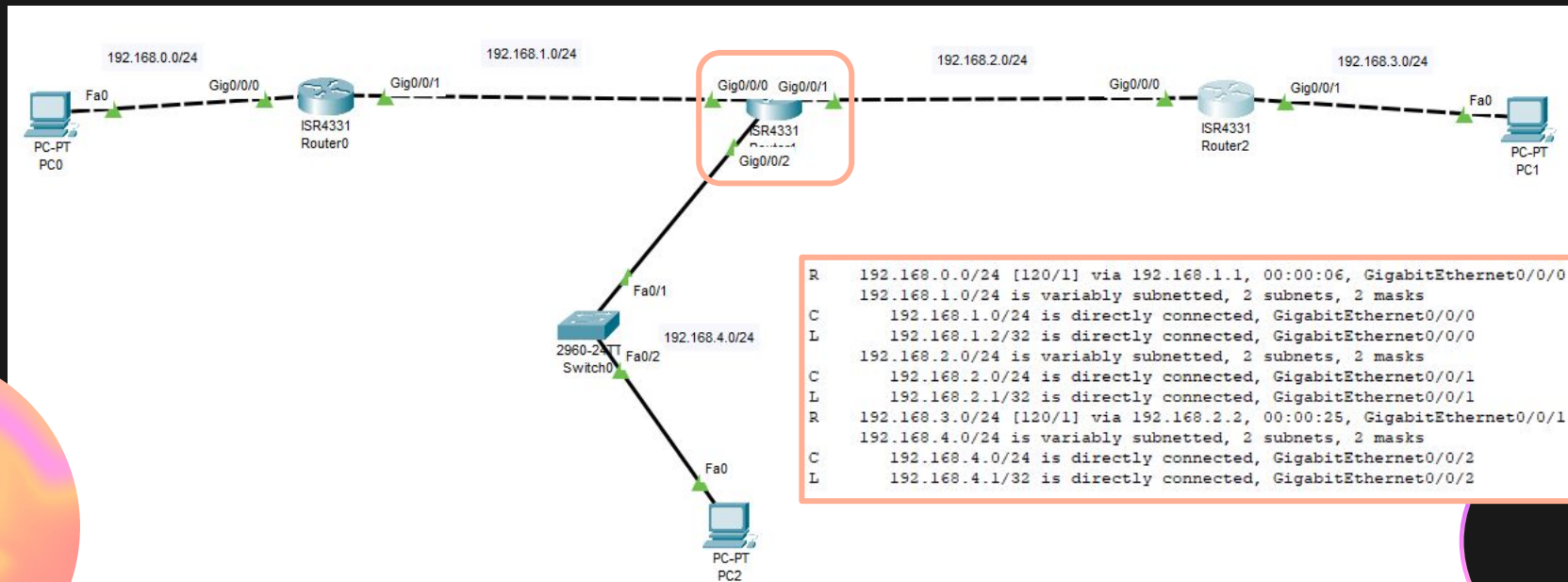


*Foi criada a tabela de roteamento dessa topologia utilizando o **RIP** de forma que todos os dispositivos consigam se conectar entre si.*

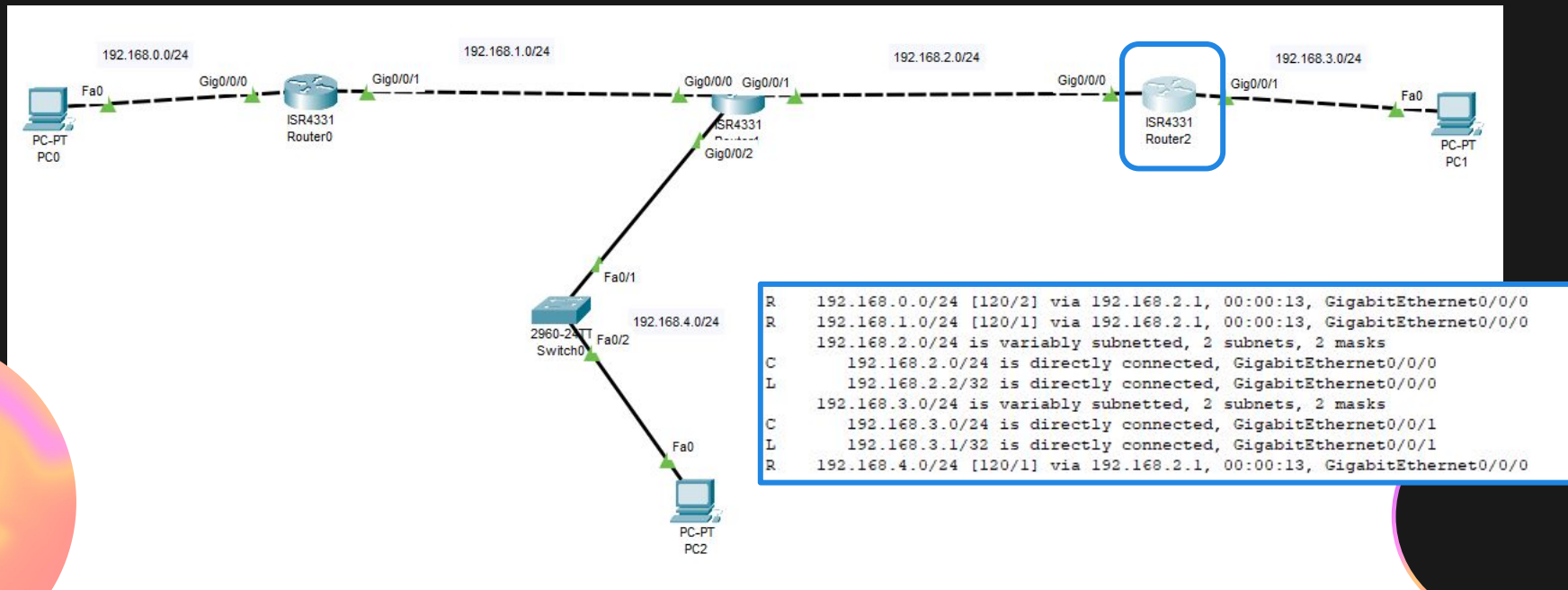
Topologia



Topologia



Topologia



Análise da Transmissão



Taxa

8ms para enviar um pacote de 32 bytes
do PC0 para o PC1 e receber a resposta
de volta ao PC0, usando protocolo RIP

The background is a dark blue gradient with several large, semi-transparent circles in shades of orange, yellow, and green. There are also several thin, white-outlined circles scattered across the image.

02

Protocolo BGP

Link para Download do
arquivo .pkt

Configurações

Topologia

<i>3 roteadores</i>	<i>3 Computadores</i>
---------------------	-----------------------

Foram criadas 6 redes

<i>192.168.0.0</i>	<i>192.168.1.0</i>	<i>192.168.2.0</i>	<i>192.168.3.0</i>	<i>192.168.4.0</i>	<i>192.168.5.0</i>
--------------------	--------------------	--------------------	--------------------	--------------------	--------------------

3 Sistemas Autônomos

<i>AS-171</i>	<i>AS-180</i>	<i>AS-200</i>
---------------	---------------	---------------

- é um protocolo de roteamento responsável por trocar informações de roteamento entre diferentes sistemas autônomos (AS)

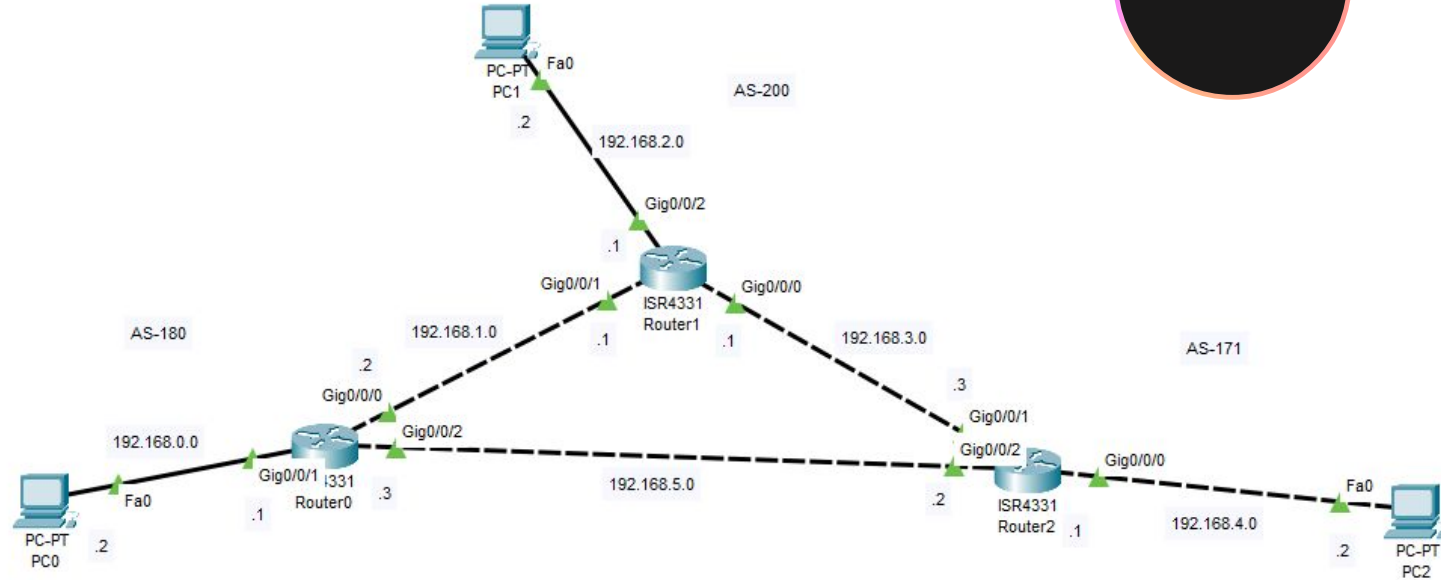
Configurações

```
Router#show ip protocols
Routing Protocol is "bgp 200"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
  Neighbor(s):
    Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
    192.168.5.3
    192.168.3.1
  Maximum path: 1
  Routing Information Sources:
    Gateway          Distance    Last Update
  Distance: external 20 internal 200 local 200
```

<< Show IP Protocols no roteador 2

Os vizinhos estão sendo apresentados (Route0 e Route1)

Topologia



Topologia

Vis.	Time(sec)	Last Device	At Device	Type
	0.011	Router1	Router0	ICMP
	0.014	Router0	PC0	ICMP
	1.000	--	Router2	TCP
	1.003	Router2	Router0	TCP
	1.004	Router0	Router2	TCP
	1.004	--	Router2	BGP
	1.006	Router2	Router0	TCP
	1.006	--	Router2	BGP
	1.006	--	Router0	BGP
	1.007	Router0	Router2	BGP
	1.007	--	Router2	BGP
	1.008	Router2	Router0	BGP
	1.008	--	Router2	BGP
	1.008	--	Router0	BGP
	1.008	--	Router0	TCP
	1.009	Router0	Router2	BGP
	1.009	--	Router0	TCP
	1.009	--	Router2	TCP
	1.011	Router2	Router0	BGP

Log do Ping entre o Router2 e o Router0 usando o protocolo BGP

Análise da Transmissão



Taxa

7ms para enviar um pacote de 32 bytes
do Router2 para o Router0, usando
protocolo BGP

Detalhe das Configurações

192.168.0.0
PC0 - 192.168.0.2
Router0 - 192.168.0.1
AS-180

192.168.2.0
PC1 - 192.168.2.2
Router1 - 192.168.2.1
AS-171

192.168.4.0
PC2 - 192.168.4.2
Router2 - 192.168.4.1
AS-200

192.168.1.0
Router0 - 192.168.1.2
Router1 - 192.168.1.1

192.168.3.0
Router1 - 192.168.3.1
Router2 - 192.168.3.3

192.168.5.0
Router2 - 192.168.5.2
Router0 - 192.168.5.3

Protocol
o BGP

192.168.0.0
PC0 - 192.168.0.2
Router0 - 192.168.0.1

192.168.1.0
Router0 - 192.168.1.2
Router1 - 192.168.1.1

192.168.2.0
Router1 - 192.168.2.1
Router1 - 192.168.2.2

192.168.3.0
Router2 - 192.168.3.1
PC1 - 192.168.3.3

192.168.4.0
PC2 - 192.168.4.2
Router2 - 192.168.4.1

Protocol
o RIP